LOWER-SECTION POLARITY AND BOOLEAN SUBSTITUTION

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Abstract. Cary's Conjecture 5.3 asserts that composing a monotone intersecting polar Boolean function with pairwise uncorrelated monotone polar Boolean functions of nonpositive mean preserves polarity. A dimension-free substitution theorem is established: if every lower section of a monotone Boolean outer function is polar, then its composition with monotone polar Boolean functions of nonpositive mean on disjoint coordinate blocks is polar. A lower section fixes an arbitrary set of inputs to -1. The proof replaces each inner function by a convex mixture of coordinate functions and the constant -1; the difference is the codifferential of a nonnegative cochain vanishing at the bottom. Multiaffine telescoping preserves this difference through the outer function. Consequently, Cary's Conjecture 4.1 implies Conjecture 5.3. Keevash's Theorem 1.1 on Kahn's flow conjecture yields Conjecture 4.1, including the nonmaximal case, and hence establishes Conjecture 5.3 in full generality.

Introduction

Intersecting families are a central subject of extremal set theory. A family $\mathcal{I} \subseteq 2^{[n]}$ is *intersecting* if any two of its members have nonempty intersection. A family $\mathcal{D} \subseteq 2^{[n]}$ is a *downset* if it is closed under taking subsets. Chvátal's conjecture, recently proved by Chang, Liu, and Liu [1], states that every finite downset admits a maximum-size intersecting subfamily which is a full star

$$\{A \in \mathcal{D} : i \in A\}$$

for some $i \in [n]$. Kleitman's strengthening and Kahn's flow conjecture place these extremal questions in the setting of weighted inequalities and upward transport on the Boolean cube [2, 3].

Cary [2] studies these questions using cochains on the discrete cube and introduces several structural conjectures concerning polarity. A function is called *polar* if its centered version is the codifferential of a cochain whose componentwise minima occur on the edges adjacent to the bottom vertex. The precise definitions and normalizations are recalled in Section 1. For Boolean functions taking values in $\{-1, 1\}$, the term *intersecting* refers to the family of subsets on which the function equals 1.

One of Cary's conjectures concerns the preservation of polarity under composition of Boolean functions. Its precise formulation is the following.

Conjecture 5.3 (Cary [2, p. 22]**).** Let

$$f : \{-1, 1\}^m \longrightarrow \{-1, 1\}, \qquad g_1, \dots, g_m : \{-1, 1\}^n \longrightarrow \{-1, 1\}$$

be monotone and polar Boolean functions. Suppose that f is intersecting, that

$$\mathbb{E} g_i \leq 0 \qquad (1 \leq i \leq m),$$

and that

$$\mathbb{E}(g_i g_j) = (\mathbb{E} g_i)(\mathbb{E} g_j) \qquad \text{whenever } i \neq j.$$

Then the composition

$$x \longmapsto f(g_1(x), \dots, g_m(x))$$

is polar.

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The condition on the inner functions is precisely vanishing pairwise covariance. They are not assumed to depend on disjoint coordinate sets; for monotone functions, this disjointness follows from the equality case of the Harris–Kleitman inequality.

The main result of this note is a dimension-free substitution theorem involving the polar lower sections of the outer function. It establishes the implication from Cary's Conjecture 4.1, which asserts polarity of every monotone intersecting Boolean function, to Conjecture 5.3. Keevash's proof of Kahn's flow conjecture [3] supplies the external result needed to establish Conjecture 4.1, including the case of nonmaximal intersecting families. Consequently, Conjecture 5.3 follows in full generality.

1. Conventions and the substitution theorem

Use $x_i \in \{-1, 1\}$, with $x_i = 1$ meaning that element i is present. All expectations in this note are uniform unless specified otherwise. Write $\mathbf{-1}$ for the bottom vertex. A 1-cochain P consists of components $P_j(x_{-j})$, and Cary's codifferential is

$$\delta P(x) = \sum_j x_j P_j(x_{-j}).$$

It has mean zero. A real cube function h is *polar* if there is such a cochain with

$$\delta P = h - \mathbb{E}h, \qquad P_j(x_{-j}) \geq P_j(\mathbf{-1}) \quad \text{for all } j, x. \tag{1}$$

For $f : \{-1, 1\}^m \to \{-1, 1\}$ and $S \subseteq [m]$, its *lower section* f_S is obtained by fixing $x_i = -1$ for $i \in S$. The section $S = \emptyset$ is f itself; constant sections are allowed. Cube functions are extended multilinearly when evaluated on $[-1, 1]^m$.

Theorem 1.1 (Lower-section substitution)**.** *Let $f : \{-1, 1\}^m \to \{-1, 1\}$ be monotone and suppose every lower section f_S is polar. Let $g_i : \{-1, 1\}^{A_i} \to \{-1, 1\}$, $1 \leq i \leq m$, be monotone and polar, where the A_i are pairwise disjoint, and suppose $b_i = \mathbb{E}g_i \leq 0$. Then $f(g_1, \ldots, g_m)$ is polar. There is no bound on m or on any $|A_i|$.*

The theorem is proved in Sections 2–3 without any external result about intersecting families. Its application to Cary's conjecture, using Keevash's theorem as an explicitly identified input, is given in Section 4.

2. Polar witnesses and affine replacements

The following lemmas isolate the properties of polar cochains needed for substitution. In particular, they allow an inner Boolean function to be replaced by an affine combination, with the residual represented by a nonnegative cochain vanishing at the bottom.

Lemma 2.1 (Nonnegativity)**.** *If h is monotone and P satisfies* (1)*, then $P_j \geq 0$ for every j.*

Proof. Put $a_j = P_j(\mathbf{-1})$. Comparing δP at the empty set and at the singleton $\{k\}$ gives

$$2a_k = h(\{k\}) - h(\emptyset) + \sum_{j \neq k} (P_j(\{k\}) - a_j) \geq 0.$$

Both monotonicity and the minimum property are used. Thus $P_j \geq a_j \geq 0$. □

Lemma 2.2 (Localization)**.** *A monotone polar function admits a polar witness supported on its essential coordinate set and depending only on those coordinates.*

Proof. Take a polar witness P. For an inessential direction j,

$$\mathbb{E}_{x_{-j}} P_j = \mathbb{E}[x_j \delta P] = \mathbb{E}[x_j (h - \mathbb{E}h)] = 0.$$

Lemma 2.1 gives $P_j \equiv 0$. Set all inessential coordinates to -1 in the remaining components. The codifferential identity and the bottom minimum property persist, proving the claim. □

Lemma 2.3 (Affine replacement)**.** *Let $g : \{-1,1\}^A \to \{-1,1\}$ be monotone and polar with $b = \mathbb{E}g \le 0$. For a polar witness ψ, put*

$$a_j = \psi_j(-\mathbf{1}), \qquad H_j = \psi_j - a_j \quad (j \in A).$$

Then

$$a_j \ge 0, \qquad \sum_{j\in A} a_j = 1 + b \le 1, \qquad H_j \ge 0, \quad H_j(-\mathbf{1}) = 0. \tag{2}$$

The affine function

$$r = b + \sum_{j\in A} a_j x_j = (-b)(-1) + \sum_{j\in A} a_j x_j \tag{3}$$

is a convex combination of -1 and coordinate functions, and

$$\delta H = g - r, \qquad \mathbb{E}r = b, \qquad r(-\mathbf{1}) = -1, \qquad -1 \le r \le 1. \tag{4}$$

For $g \equiv -1$ use the zero witness and $r = -1$.

Proof. Since g is monotone Boolean and $b \le 0$, its bottom value is -1. Evaluating $\delta\psi = g - b$ there gives $-\sum_j a_j = -1 - b$. Nonnegativity follows from Lemma 2.1. The weights in (3) sum to $-b + (1 + b) = 1$ and are nonnegative. Finally

$$\delta H = g - b - \sum_j a_j x_j = g - r.$$

All other assertions follow directly. □

Lemma 2.4 (Two closure operations)**.** *Convex combinations of polar functions are polar. If G is polar and $R_j \ge 0$ with $R_j(-\mathbf{1}) = 0$ for every component, then $G + \delta R$ is polar.*

Proof. For the first assertion take the same convex combination of witnesses. For the second, if Ω witnesses G, then $\Omega + R$ has the correct codifferential because $\mathbb{E}\delta R = 0$, and

$$(\Omega_j + R_j)(x) \ge \Omega_j(-\mathbf{1}) = (\Omega_j + R_j)(-\mathbf{1}).$$

□

3. Proof of the dimension-free substitution theorem

The proof separates the composition into an affine part and a nonlinear remainder. The lower-section hypothesis provides a polar witness for the affine part, while monotonicity allows the remainder to be represented by a nonnegative cochain with zero bottom values.

Apply Lemma 2.3 on each block. Write

$$r_i = b_i + \sum_{j\in A_i} a_{ij} x_j, \qquad \delta H^{(i)} = g_i - r_i.$$

In particular $H_j^{(i)} \ge 0$ and $H_j^{(i)}(-\mathbf{1}) = 0$. Set

$$F = f(g_1, \dots, g_m), \qquad G = f(r_1, \dots, r_m).$$

The affine composition is a mixture of lower sections. Independently for each i, choose $\sigma_i \in \{*\} \cup A_i$ with

$$\Pr(\sigma_i = *) = -b_i, \qquad \Pr(\sigma_i = j) = a_{ij},$$

and let $h_i^\sigma = -1$ if $\sigma_i = *$ and $h_i^\sigma = x_{\sigma_i}$ otherwise. The multilinearity of f gives the pointwise identity

$$G(x) = \mathbb{E}_\sigma f(h_1^\sigma(x), \dots, h_m^\sigma(x)). \tag{5}$$

This expectation is over the auxiliary choices, with x held fixed. Every function inside the expectation is a lower section of f, followed by an injective renaming of its remaining coordinates and addition of dummy variables. Injectivity uses the disjointness of the blocks. By hypothesis each function is polar, including constant sections. Choose and average their embedded witnesses. The result is a polar witness Ω with

$$\delta\Omega = G - \mathbb{E}G. \tag{6}$$

There are only finitely many choices, so this is an ordinary finite convex combination.

The nonlinear remainder is a nonnegative bottom-zero flow. Define

$$C_i(x) = \partial_i f\big(g_1(x), \dots, g_{i-1}(x), 0, r_{i+1}(x), \dots, r_m(x)\big), \tag{7}$$

where $\partial_i f = \frac{1}{2}(f|_{x_i=1} - f|_{x_i=-1})$ and its own argument is irrelevant. This coefficient depends only on blocks other than A_i. Moreover $C_i \geq 0$: the derivative is nonnegative on Boolean vertices by monotonicity, and its multilinear extension is a convex combination of those vertex values throughout $[-1,1]^{m-1}$. All the inputs in (7) lie in that box.

Replace r_1 by g_1, then r_2 by g_2, and so on. The resulting exact telescoping identity is

$$F - G = \sum_{i=1}^{m} C_i(g_i - r_i). \tag{8}$$

For $j \in A_i$ define

$$R_j = C_i H_j^{(i)}. \tag{9}$$

This is a legitimate component: C_i ignores the whole of A_i, and $H_j^{(i)}$ ignores x_j. Hence

$$\delta R = \sum_i C_i \delta H^{(i)} = \sum_i C_i(g_i - r_i) = F - G. \tag{10}$$

Every R_j is nonnegative and vanishes at the bottom. Thus $\Phi = \Omega + R$ is polar by Lemma 2.4. Equations (6)–(10) give

$$\delta\Phi = F - \mathbb{E}F.$$

For completeness, disjoint blocks and multilinearity also give $\mathbb{E}F = \mathbb{E}G = f(b_1, \dots, b_m)$. This proves Theorem 1.1. □

Remark 3.1 (Explicit output witness)**.** *If Ω^σ is the embedded witness of the section in* (5)*, and w_σ is its product weight, the construction is*

$$\boxed{\Phi_j = \sum_\sigma w_\sigma \Omega_j^\sigma + C_i(\psi_j^{(i)} - a_{ij}), \qquad j \in A_i.}$$

Remark 3.2 (Sections of positive weight)**.** *For fixed inner means $b_1, \dots, b_m$, the mixture in* (5) *only requires polarity of those lower sections f_S for which*

$$\{i : b_i = -1\} \subseteq S \subseteq \{i : b_i < 0\}.$$

These are exactly the sets of coordinates that can be fixed to -1 with positive probability in the mixture.

In particular, when every $b_i = 0$, only $S = \emptyset$ occurs. In this balanced case, polarity of the outer function f alone suffices, and the construction recovers Cary's Theorem 4.5 after the disjointness of essential coordinate supports is deduced from the zero-covariance condition. For negative inner means, the additional lower sections become relevant.

4. Application to Cary's Conjecture 5.3

The application proceeds in two steps. First, Keevash's flow theorem is used to establish Cary's Conjecture 4.1, including the nonmaximal case. The lower-section substitution theorem then yields Conjecture 5.3.

The external input and the nonmaximal case. We use [3, Theorem 1.1] as an external result, without reproducing its spectral proof. For a monotone odd real function h, choose, for every nonempty $T \subseteq [n]$ and $j \in T$,

$$\lambda_T(j) = \frac{\widehat{h}(T)^2}{|T|}, \qquad \lambda_j = \sum_{T \ni j} \lambda_T(j).$$

The Fourier coefficients here are taken in the $\{-1,1\}$ convention. Keevash's convention differs by a sign depending on $|T|$, which disappears after squaring.

Since h is odd, its constant Fourier coefficient vanishes. Thus Parseval gives

$$\lambda_j \geq 0, \qquad \sum_j \lambda_j = \sum_{T \neq \emptyset} \widehat{h}(T)^2 = \mathbb{E}h^2.$$

Writing $z_j = (1 + x_j)/2$ and $L = \sum_j \lambda_j z_j$, Keevash's theorem provides an upward transport from L to h_+^2.

Proposition 4.1 (Consequence of Keevash's theorem)**.** *Every monotone intersecting Boolean function is polar, including the nonmaximal case of Cary's Conjecture 4.1.*

Proof. For its winning family $\mathcal{F}$, put $q = \mathbf{1}_{\mathcal{F}}$ and $h(x) = q(x) - q(-x)$. Then h is monotone and odd. Intersection gives $q(x)q(-x) = 0$, so $h_+^2 = q$ and $\mathbb{E}h^2 = 2\mathbb{E}q$. By Parseval and the preceding identities,

$$\sum_j \lambda_j = \mathbb{E}h^2 = 2\mathbb{E}q, \qquad \mathbb{E}L = \mathbb{E}q.$$

The upward transport supplied by Keevash consists of nonnegative coefficients $\beta(A, B)$, supported on $A \subseteq B$, with prescribed marginals

$$\sum_{B \supseteq A} \beta(A, B) = L(A), \qquad \sum_{A \subseteq B} \beta(A, B) = q(B).$$

For each pair $A \subseteq B$, route the mass $\beta(A, B)$ along an increasing path from A to B; for $A = B$ the path is empty. Summing these path contributions gives a nonnegative edge cochain ρ satisfying, at every vertex v,

$$(\delta\rho)(v) = \sum_{A \subseteq B} \beta(A, B)\big(\mathbf{1}_{\{B\}}(v) - \mathbf{1}_{\{A\}}(v)\big) = q(v) - L(v).$$

Both q and L vanish at $\boldsymbol{-1}$. Hence

$$0 = (\delta\rho)(\boldsymbol{-1}) = -\sum_j \rho_j(\boldsymbol{-1}).$$

Since ρ is nonnegative, every $\rho_j(\boldsymbol{-1})$ must vanish. Now

$$\Xi_j = \lambda_j + 2\rho_j$$

has bottom minima and

$$\delta\Xi = 2q - \sum_j \lambda_j = 2q - 2\mathbb{E}q = f - \mathbb{E}f.$$

The empty winning family uses the zero witness. □

Corollary 4.2 (The logical implication)**.** *Cary's Conjecture 4.1 implies his Conjecture 5.3. Consequently Conjecture 5.3 follows in full generality from* [3, Theorem 1.1] *and Theorem 1.1.*

Proof. Let $f, g_1, \ldots, g_m$ satisfy the hypotheses of Cary's Conjecture 5.3 as stated in [2, p. 22]. Explicitly, the hypotheses include $b_i = \mathbb{E}g_i \leq 0$, $\mathbb{E}(g_i g_k) = b_i b_k$ for $i \neq k$, and monotonicity and polarity of $f, g_1, \ldots, g_m$, with f additionally intersecting.

Covariance zero between monotone functions implies disjoint essential supports, by the equality case of Harris–Kleitman. One can see this directly: conditioning on any coordinate j gives

$$\operatorname{Cov}(g_i, g_k) = \tfrac{1}{2}\operatorname{Cov}(g_i^-, g_k^-) + \tfrac{1}{2}\operatorname{Cov}(g_i^+, g_k^+) + \tfrac{1}{4}(\mathbb{E}g_i^+ - \mathbb{E}g_i^-)(\mathbb{E}g_k^+ - \mathbb{E}g_k^-).$$

The conditional covariances are nonnegative by Harris–Kleitman. Alternatively, their nonnegativity follows by induction on the number of coordinates, applying the same identity to the conditional functions.

For a monotone function, a coordinate j is essential precisely when the nonnegative difference $g_i^+ - g_i^-$ is not identically zero. In that case its expectation is strictly positive. Thus a coordinate essential for both g_i and g_k makes the last term strictly positive, contradicting $\operatorname{Cov}(g_i, g_k) = 0$.

Consequently the essential supports of the inner functions are pairwise disjoint. Localize their polar witnesses by Lemma 2.2.

Every lower section of the monotone intersecting f is again monotone intersecting, or the constant -1. Under Conjecture 4.1 every such section is polar. Apply Theorem 1.1 to obtain the conclusion of 5.3. Proposition 4.1 supplies Conjecture 4.1 from the external theorem, proving the second assertion. □

Remark 4.3 (Scope of the conclusion)**.** *The inner functions need not be intersecting, and the output need not be intersecting either. The hypothesis that the outer f is polar is part of Cary's original statement; under Proposition 4.1 it is automatic from monotonicity and intersection. No dimension bound or threshold representation is imposed.*

5. An exact nonintersecting-output check

The general proof is the mathematical argument of Sections 2–4, not an extrapolation from finite computations. The supplementary program `verify_examples.py`, written independently for the mathematical audit of this note, checks the construction using exact integer and rational arithmetic. Its output is supplied in `results.json`.

The program uses only the Python standard library and can be executed with Python 3.10 or later:

```
python3 verify_examples.py > results.json
```

The program uses Python assertions for its checks and must therefore be run without the `-O` optimization option.

The outer function is the Fano upset with minimal winners

$$123, \quad 145, \quad 167, \quad 246, \quad 257, \quad 347, \quad 356.$$

The verifier constructs rational polar witnesses for all 128 lower sections by an exact integer-capacity flow computation. It checks $3^7 = 2187$ codifferential identities and the bottom-minimum conditions directly. No previously recorded certificate is required as input.

One inner function, on four coordinates, has minimal winners $12, 23, 34$. It is balanced and not intersecting. In terms of $z_j = (1 + x_j)/2$, a polar witness is

$$\psi_1 = z_2(1 - z_3), \quad \psi_2 = \tfrac{1}{2}, \quad \psi_3 = \tfrac{1}{2}, \quad \psi_4 = (1 - z_2)z_3.$$

Its affine replacement is $(x_2 + x_3)/2$. Another inner function is the two-coordinate conjunction, with mean $-1/2$, witness $\psi_1 = z_2,\ \psi_2 = 1/2$, and replacement $-1/2 + x_2/2$. Coordinates and the constant -1 supply the other blocks.

Check	Dimension	Vertices	Mixture terms	Mean
All coordinate inputs	7	128	1	0
Path, conjunction, five coordinates	11	2048	4	$-9/64$
Path, two conjunctions, four coordinates	12	4096	8	$-9/32$
Conjunction, path, constant, four coordinates	10	1024	4	$-27/64$

The program verifies the inner affine replacements, the exact mixture identity, the telescoping divergence, nonnegativity and bottom-zero values of the remainder, the final minima, own-coordinate independence, and correct centering at all 7296 composition vertices.

In the eleven-coordinate example, use the path on $\{1, 2, 3, 4\}$, the conjunction on $\{5, 6\}$, and coordinates $7, 8, 9, 10, 11$ for the five remaining Fano inputs. The two disjoint sets

$$\{1, 2, 5, 6, 7\}, \qquad \{3, 4, 8, 9\}$$

are both winning: their outer patterns are 123 and 145. Thus the output is not intersecting, yet the constructed witness is polar. Merely applying ordinary polarity directly to the output would not justify this example.

6. Scope and concluding remarks

The new implication is

$$\boxed{\text{Cary 4.1} \implies \text{Cary 5.3.}}$$

Combined with Keevash's Theorem 1.1, it yields Conjecture 5.3 in every finite dimension.

There is a useful check on the sign restriction. For a nonconstant monotone Boolean polar g, write $b = \mathbb{E}g$ and $a_j = \psi_j(-\mathbf{1})$. Evaluation at the two poles gives

$$1 + b = \sum_j a_j \le \sum_j \psi_j(\mathbf{1}) = 1 - b,$$

so $b \le 0$ automatically. Among monotone Boolean polar functions, the explicit nonpositive-mean assumption therefore serves to exclude the constant $+1$. This exclusion matters: take the outer three-variable majority and inner functions $x_1, x_2, +1$. All are monotone and polar, and the inner covariances vanish, but the output is the two-variable OR. Its mean is $1/2$, so it cannot be polar by the preceding pole comparison. This is outside the stated conjecture, not a counterexample to it.

The conclusion concerning Cary's Conjecture 5.3 uses Keevash's Theorem 1.1 as an external result. The lower-section substitution theorem is proved independently of that theorem. No new proof of Kahn's flow conjecture is claimed.

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